

Contextual Bandit-based MPC Tuning for Optimized Personalization of Physical Activity Behavioral Interventions[★]

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Abstract: “Just-in-Time” adaptive interventions (JITAI) can provide daily step goals and financial incentives (e.g., reward points) to enable participants to reach healthy levels of physical activity. However, personalization and optimization remain challenging because participant responses vary over time and between individuals. Recent JITAI efforts have benefited from control-oriented strategies using system identification and Model Predictive Control (MPC), with *YourMove*(NCT05598996) providing an illustrative example. MPC offers a structured decision framework for selecting feasible goal trajectories from theory-guided dynamic behavioral models, but its closed-loop performance depends on tuning choices that trade off tracking speed, smoothness, constraint satisfaction, and other criteria, making manual selection difficult. A contextual multi-armed bandit formulation can instead use a reward function to select MPC tuning parameters that best align the controller with the intervention outcomes of interest, conditioned on observed participant context. This paper presents a simulation-based proof of concept in which a TD3-based agent is used as an outer-loop tuner that learns the optimal MPC tuning parameters, while MPC produces feasible responses that satisfy intervention constraints. Results demonstrate improved closed-loop behavior tracking and increased self-efficacy relative to a fixed-tuning MPC baseline, and multi-seed training indicates reproducible convergence to high-reward tuning policies. Both novel and practical insights on the synergism of machine learning and control engineering are outcomes of this research.

Keywords: mHealth; behavioral interventions; model predictive control; contextual bandits; Social Cognitive Theory

1. INTRODUCTION

Physical activity offers well-documented benefits to individual and public health, but many adults still do not achieve recommended activity levels. Epidemiological evidence indicates that increasing daily steps from approximately 4,000 to 8,000 per day is associated with a substantial reduction in all-cause mortality risk, while sedentary behavior remains prevalent across multiple countries (Department for Digital, Culture, Media & Sport, 2017; Centers for Disease Control and Prevention, 2021). This persistent gap motivates scalable, adaptive intervention strategies that personalize recommendations in real time rather

than relying on static recommendations. Recent advances in smartphones and wearable sensors have enabled continuous measurement of physical activity, making real-time, data-driven interventions increasingly feasible. Based on this, the *YourMove* intervention (NIH R01CA244777; ClinicalTrials.gov NCT05598996) has demonstrated how Model Predictive Control (MPC) using a behavioral model grounded in social cognitive theory (SCT) identified from participant data (Martín et al., 2020) can deliver personalized, adaptive daily step-count goals and financial incentives (“*expected points*”, EP; El Mistiri et al. (2025a,b)).

Despite its advantages, MPC performance depends critically on tuning choices such as move suppression weights, reference targets, and output weights, which are difficult to optimize manually and inherently participant dependent. Move-suppression weights, when allowed to vary over the move horizon, offer a valuable additional degree of freedom

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but tuning such profiles is tedious, so traditionally these are set to a single fixed value. This motivates pursuing data-driven strategies. Using RL alone, however, lacks interpretability and is sample intensive; furthermore, hard constraints necessary for safety cannot be enforced without additional structure (Bloor et al., 2025).

Combining RL with MPC addresses both limitations: MPC provides the structured decision layer that enforces constraints and maintains feasibility, while RL brings data-driven adaptability by tuning MPC parameters from closed-loop behavioral trajectories. This paper presents a simulation-based “proof of concept” in which a TD3-based agent is used as an outer-loop contextual bandit optimizer that searches over MPC parameters, including horizon-varying move-suppression weights and financial incentive reference levels, based on closed-loop behavioral trajectories, while MPC achieves constraint enforcement and structured decision-making. The framework is evaluated by how effectively the tuned controller drives the behavior of the daily steps of participants towards the desired activity target while simultaneously promoting high motivational state trajectories under closed-loop constraints while examining how the reward formulation shapes the learned tuning and closed-loop behavior. Results provide preliminary evidence that learning-based tuning can promote intervention goals (e.g., step counts, high self-efficacy) alongside fast setpoint tracking and constraint satisfaction. Reliance on manual retuning when deploying to new participants or adapting to behavioral changes is reduced. This idea has also shown promise in other domains, where RL has been used to adjust MPC weight coefficients while maintaining safety constraints (Wang et al., 2023) and deep RL has been used to tune MPC cost weights and horizons to improve tracking performance relative to fixed parameter baselines (Hamadeh et al., 2025). The key contribution of this paper is a contextual bandit formulation for data-driven tuning of MPC parameters. In particular, it tunes horizon-varying weights, a design choice that is difficult to optimize, while improving closed-loop performance under the same MPC constraints.

The paper is structured as follows. Section 2 presents the SCT-based behavioral model and the baseline MPC formulation from *YourMove*. Section 3 describes the proposed contextual bandit+MPC framework, including the reward function and TD3-based learning setup. Section 4 reports simulation results and discusses their implications. Finally, Section 5 concludes with a summary and future directions.

2. SYSTEM MODEL AND MPC FORMULATION

2.1 SCT-based OLSE model

The behavioral dynamics follow a control-oriented Social Cognitive Theory (SCT) fluid analogy model in which key SCT constructs are represented as interacting inventories and flows (Martín et al., 2020). The SCT subsystem considered includes Self Efficacy (SE), Behavior (daily steps), and Behavioral Outcomes (BehOut), and is commonly referred to as an Operant Learning Self Efficacy (OLSE) system (Fig. 1, El Mistiri et al. (2025b)). In this paper, the OLSE model is simulated deterministically, although stochastic disturbances and measurement noise can be

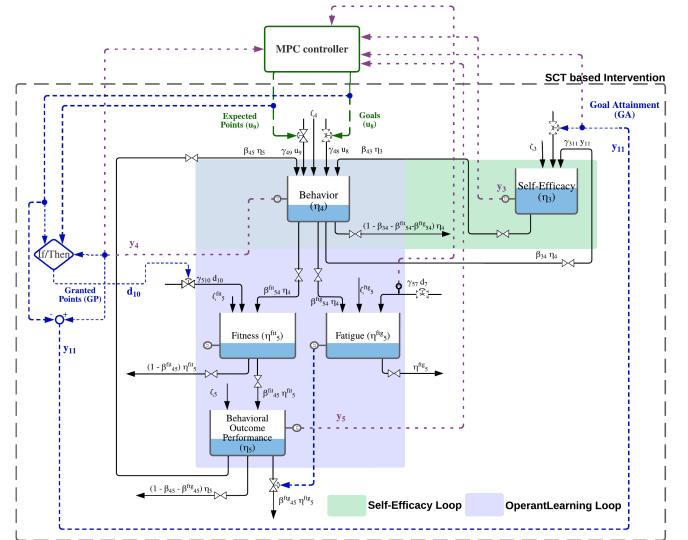


Fig. 1. Schematic of the fluid-analogy SCT model for the OLSE intervention system with dual behavioral outcomes and MPC control. Adapted from (Martín et al., 2020; El Mistiri et al., 2025b).

incorporated when desired. Self-efficacy (SE) is a core construct in SCT that represents the individual’s perceived capability to engage in the target behavior. Its dynamics depend on other key SCT constructs, including behavior, and on other inflows and outflows in the system. Behavioral Outcomes represent the physical and psychological consequences of engaging in the behavior (steps/day), such as fatigue and positive reinforcement. In the model, Behavioral Outcomes is split into dual components to capture fitness-related and fatigue-related effects, while permitting influence from external factors such as environmental context and incentives. In *YourMove*, the manipulated inputs are the daily step goals and the expected points incentive (El Mistiri et al., 2025a,b). Expected points (EP) denote the planned incentive points communicated to the participant for a given day, while granted points (GP) denote the realized points awarded based on whether the participant meets the prescribed goal. Granted points can then be accumulated and redeemed for a financial reward at the end of the intervention. Goal Attainment (GA) is computed as the deviation between behavior and the prescribed goal and is used both for evaluation and for optional constraint enforcement.

Among these outputs, y_3 denotes SE, y_4 denotes behavior (steps per day), and y_5 denotes behavioral outcomes, with dual dynamics captured through fitness $y_{5,fit}$ and fatigue $y_{5,ftg}$. The manipulated inputs are $u_8(t)$ for daily goals (steps per day) and $u_9(t)$ for expected points (points per day), while measured disturbances include granted points $\xi_{10}(t)$ and environmental context $\xi_7(t)$. Granted points are awarded when the measured behavior meets or exceeds the prescribed daily step goal, with the granted points equal to the expected points value for that day when the goal is met and equal to zero otherwise.

The continuous time OLSE subsystem is represented in lumped parameter form with the following state, input, disturbance, and output definitions:

$$x(t) = \begin{bmatrix} \eta_3(t) \\ \eta_4(t) \\ \eta_{5,\text{fit}}(t) \\ \eta_{5,\text{ftg}}(t) \\ \eta_5(t) \end{bmatrix}, \quad u(t) = \begin{bmatrix} u_8(t) \\ u_9(t) \end{bmatrix}, \quad (1)$$

$$d(t) = \begin{bmatrix} \xi_{10}(t) \\ \xi_7(t) \end{bmatrix}, \quad y(t) = \begin{bmatrix} y_3(t) \\ y_4(t) \\ y_{5,\text{fit}}(t) \\ y_{5,\text{ftg}}(t) \\ y_5(t) \\ y_{11}(t) = y_4(t) - u_8(t) \end{bmatrix} \quad (2)$$

MATLAB/Simulink is used to implement the simplified SCT model and MPC controller shown in Fig. 1. The controller is designed from the discrete-time state-space representation of the OLSE subsystem reported in El Mistiri et al. (2025b), and the variable notation used throughout this paper follows the same convention to support consistency, reproducibility, and ease of cross-reference. For the simulation studies in this paper, the OLSE gains and time constants are fixed to emulate an adherent participant, which means a participant whose behavior responds positively to appropriately selected step goals and incentives. The parameter values used are:

- Time constants: $\tau_3 = 1$, $\tau_4 = 2$, $\tau_5 = 5$.
- Gains: $\gamma_{311} = 1.3$, $\gamma_{48} = 1$, $\gamma_{49} = 0.3$, $\gamma_{57} = 4$, $\gamma_{510} = 5$.
- Coupling coefficients: $\beta_{34} = 0.5$, $\beta_{43} = 0.2$, $\beta_{45} = 0.2$, $\beta_{54} = 0.4$.
- Dual behavioral-outcome dynamics (fitness/fatigue):
 $\tau_5^{\text{fit}} = 2$, $\tau_5^{\text{ftg}} = 1$, $\tau_5 \approx 0$,
 $\beta_{45}^{\text{fit}} = 0.5$, $\beta_{45}^{\text{ftg}} = 0.2$, $\beta_{45} = 0.2$,
 $\beta_{54}^{\text{fit}} = 0.8$, $\beta_{54}^{\text{ftg}} = 0.7$, with $\gamma_{57} = 4$ and $\gamma_{510} = 5$.

In *YourMove*, model parameters are estimated from participant data using prediction-error methods, which lead to participant-specific models (El Mistiri et al., 2025a).

2.2 Baseline MPC setup

A receding horizon MPC controller selects daily step goals and expected points by minimizing the quadratic objective in Eqn. (3) over a prediction horizon p and move horizon m (El Mistiri et al., 2025b). At each day k , the optimization selects the manipulated variables $u_{8,k}$ (daily goal) and $u_{9,k}$ (expected points) to regulate the primary controlled output, behavior $y_{4,k}$ (steps/day), toward the setpoint $r_k = 10,000$ steps/day. The predicted state and output trajectories entering the objective are generated by propagating the OLSE dynamics forward over the horizon using the planned future manipulated variables together with measured disturbances. This MPC formulation serves as the baseline controller, with identical output and manipulated-variable weights used in both cases to isolate the effect of the RL-selected horizon-varying move-suppression weights.

Eqn. (3) defines the MPC controller's objective function, which the optimizer minimizes at each day k to compute the next manipulated-variable by trading off reference tracking, operating-point preference, and move suppression. The first term penalizes output tracking error $y(k+l|k) - r(k+1)$, with $\Gamma_Y^{(l)}$ setting the relative impor-

Table 1. MPC configuration and constraints used in the closed-loop RL+MPC framework.

Setting	Value
Sampling time	$T_s = 1$ day
Prediction horizon	$p = 20$
Move horizon	$m = 5$
Behavior bounds	$y_4 \in [0, \infty)$
Output variable weights (Γ_Y)	$\{\Gamma_{y_3}, \Gamma_{y_4}, \Gamma_{y_{5,\text{fit}}}, \Gamma_{y_{5,\text{ftg}}}, \Gamma_{y_{11}}\} = [0, 1, 0, 0, 0]$
Manipulated variable weights (Γ_u)	$\{\Gamma_{u_8}, \Gamma_{u_9}\} = [0, 1]$
Move-suppression weights ($\Gamma_{\Delta u}$)	$\Gamma_{\Delta u_8} = \Gamma_{\Delta u_9} = 0$
Expected Points Reference	$u_{9r} = 500$

tance of each output at prediction step l . The second term penalizes deviation of the manipulated variables from a setpoint u_r , such as 10,000 steps, and is weighted by $\Gamma_u^{(l)}$. The third term penalizes input increments $\Delta u(k+l-1)$, where $\Gamma_{\Delta u}^{(l)}$ enforces move suppression and shapes the smoothness and aggressiveness of the manipulated variables. The MPC configuration and weights used in the closed-loop experiments are summarized in Table 1.

$$J_k = \sum_{l=1}^p \left\| \Gamma_Y^{(l)} (y(k+l|k) - r(k+1)) \right\|_2^2 + \sum_{l=1}^m \left\| \Gamma_u^{(l)} (u(k+l-1) - u_r) \right\|_2^2 + \sum_{l=1}^m \left\| \Gamma_{\Delta u}^{(l)} \Delta u(k+l-1) \right\|_2^2 \quad (3)$$

To preserve feasibility, safety, and usability, the below constraints enforces bounds on outputs, manipulated variables, and move sizes respectively,

$$y_{\min} \leq y(k+i) \leq y_{\max} \quad \forall 1 \leq i \leq p, \quad (4)$$

$$u_{\min} \leq u(k+i) \leq u_{\max} \quad \forall 0 \leq i \leq m-1, \quad (5)$$

$$\Delta u_{\min} \leq \Delta u(k+i) \leq \Delta u_{\max} \quad \forall 0 \leq i \leq m-1. \quad (6)$$

Hard bounds typically reflect physical or operational limitations, for instance the manipulated variables limits are:

$$0 \leq u_8(k+i) \leq 20,000, \quad 0 \leq i \leq m-1 \quad (7)$$

$$0 \leq u_9(k+i) \leq 500, \quad 0 \leq i \leq m-1 \quad (8)$$

and additional minimum thresholds are applied to a key motivational state variable self-efficacy as well as goal attainment that influences SE, to help prevent participant motivation from dropping too low during the intervention:

$$-15000 \leq y_3(k+i) \leq \infty \quad \forall 0 \leq i \leq p. \quad (9)$$

$$-1000 \leq y_{11}(k+i) \leq \infty \quad \forall 0 \leq i \leq p. \quad (10)$$

3. CONTEXTUAL BANDIT FRAMEWORK FOR MPC TUNING

Figure 2 summarizes the closed-loop contextual bandit MPC architecture used in this paper. A contextual bandit is a specialized, one-step type of reinforcement learning in which a single action is selected given an observed context and performance is evaluated over the resulting episode. In this setting, the SCT-based OLSE model provides daily observations of physical activity behavior and self-efficacy that define the context available to the tuning policy and serve as measured outputs for MPC prediction and constraint handling. At the start of each episode, a Twin

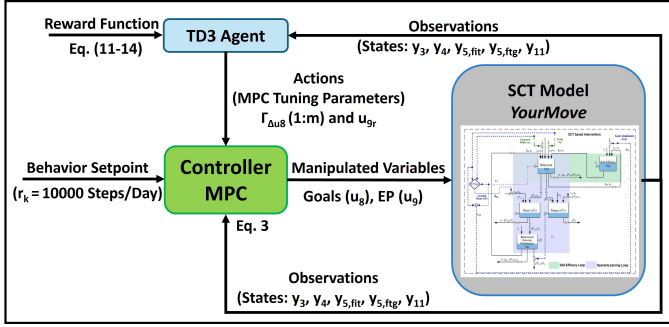


Fig. 2. Contextual bandit-tuned MPC framework. A TD3-based tuning policy selects MPC parameters once per episode, and MPC computes daily goals and expected points under constraints.

Delayed Deep Deterministic policy gradient (TD3) based agent maps the initial context to continuous MPC tuning parameters, which remain fixed throughout the episode, while MPC solves a constrained QP daily to generate the goal and expected-points inputs. Daily rewards are computed from the tracking, safety, and motivation terms in Section 3.2, and the tuning policy is updated from stored experience.

3.1 Problem formulation

The learning problem is posed as online episodic tuning of MPC design parameters in a contextual bandit setting. The environment and MPC operate at a daily sampling time ($T_s = 1$ day), and each episode spans $N_d = 15$ days. At the start of each episode, the environment is reset and the initial context O_0 is obtained from the current closed-loop state. During the episode, MPC solves its constrained QP once per day using the latest observation O_k to compute daily goals and expected-points inputs. In contrast, the tuner acts once per episode: at $k = 0$, the TD3 actor maps O_0 to a continuous tuning action $a = [\Gamma_{\Delta u_8}(1:m), u_{9r}]^T$, where $\Gamma_{\Delta u_8}(1:m)$ denotes move-horizon-varying move-suppression weights on the goal and u_{9r} is the expected-points reference. This tuning is then held fixed as the closed loop evolves over the episode and generates the observation sequence $\{O_k\}_{k=0}^{N_d}$. In the current implementation, learning is driven by the terminal reward R_{N_d} , computed at the end of the episode to summarize performance under the fixed tuning action.

Accordingly, experience is generated at the daily time scale and stored in the replay buffer $\mathcal{D}$ together with the episode-constant action a (daily records of the form (O_k, a, R_k, O_{k+1})). The replay buffer is sampled only for learning updates. Since the tuner acts only once per episode, the learning signal used for policy updates is the terminal performance measure R_{N_d} . In this bandit-style setting, TD3 (Fujimoto et al., 2018) serves as a deep value-based optimizer over the continuous tuning space: the twin critics Q_{ϕ_1} and Q_{ϕ_2} learn to predict expected terminal reward as a function of context and action, and the actor μ_θ is updated on the delayed TD3 schedule to select actions that maximize the learned critic value, with target-policy smoothing and Polyak-averaged target networks used for stabilization. This enables sample-efficient episode-level tuning from end-of-episode performance.

Algorithm 1 TD3-style optimizer for Contextual Bandit

- 1: **Input:** MPC parameters (Table 1), episode length $N_d = 15$, $T_s = 1$ day, terminal reward R_{N_d}
- 2: **Action (per episode):**
 $a = [\Gamma_{\Delta u_8}(1:m), u_{9r}]^T$, $\Gamma_{\Delta u_8}(i) \in [0, 1000]$, $u_{9r} \in [0, 500]$
- 3: **Observation (daily):**
 $O_k = \{y_3, y_4, y_5, fltr, y_5, fltr, y_{11}\}$
- 4: Initialize actor μ_θ , critics Q_{ϕ_1}, Q_{ϕ_2} , target networks, replay buffer $\mathcal{D} \leftarrow \emptyset$
- 5: **for** episode = 1, 2, ..., 3000 **do**
- 6: Reset environment; observe initial context O_0
- 7: Select episode-constant tuning action
 $a \leftarrow \text{clip}(\mu_\theta(O_0) + \epsilon)$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$
- 8: Configure MPC for the episode using a
- 9: **for** $k = 0, 1, \dots, N_d - 1$ **do**
- 10: Solve daily MPC QP (Eqn. 3) $\rightarrow (u_{8,k}, u_{9,k})$
- 11: Apply $(u_{8,k}, u_{9,k})$ to the SCT model $\rightarrow O_{k+1}$
- 12: Compute daily reward $R_k \leftarrow R(y_{4k}, y_{3k})$
- 13: Store daily experience (O_k, a, R_k, O_{k+1}) in $\mathcal{D}$
- 14: **end for**
- 15: **Off-policy TD3 updates (after the episode):**
 sample mini-batches from $\mathcal{D}$ and update critics/actor (delayed TD3), with target networks updated via Polyak averaging
- 16: **end for**

3.2 Reward function

The reward function encodes the objective of tracking the 10,000 steps/day setpoint, penalizing unsafe high-step trajectories, and encouraging a high motivational state proxy (Fig. 3). At each day k , the environment returns a scalar reward $R_k = R(y_{4k}, y_{3k})$ defined in (11)–(14). In this episode-level tuning setup, daily rewards are recorded, while tuning updates are driven by the terminal reward R_{N_d} . This reward specification is used as a preliminary design for assessing how TD3-selected MPC tuning parameters shape closed-loop tracking, safety, and motivational outcomes.

$$R(y_4, y_3) = \left(r_{\text{track}}(y_4) + r_{\text{safe}}(y_4) + r_{\text{se}}(y_3) \right), \quad (11)$$

$$r_{\text{track}}(y_4) = \begin{cases} 10000, & |r_k - y_4| \leq 1000, \\ 0, & y_4 > r_k + 1000, \\ -|r_k - y_4|, & \text{otherwise,} \end{cases} \quad (12)$$

$$r_{\text{safe}}(y_4) = \begin{cases} -2000, & y_4 > 20000, \\ 0, & \text{otherwise,} \end{cases} \quad (13)$$

$$r_{\text{se}}(y_3) = \begin{cases} 10000, & y_3 > 9000, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

4. RESULTS & DISCUSSION

This section evaluates the closed-loop contextual bandit + MPC framework in Section 3 using the observation and action mentioned in Algorithm 1. The experiment tests whether the TD3-based tuner can learn the move-horizon-varying move-suppression weights $\Gamma_{\Delta u_8}(1:m)$ and the expected-points reference u_{9r} such that MPC drives behavior y_4 toward the 10,000 steps/day setpoint while enforcing the goal-attainment lower bound constraint $y_{11} \geq -1000$, using the reward function described in Section 3.2.

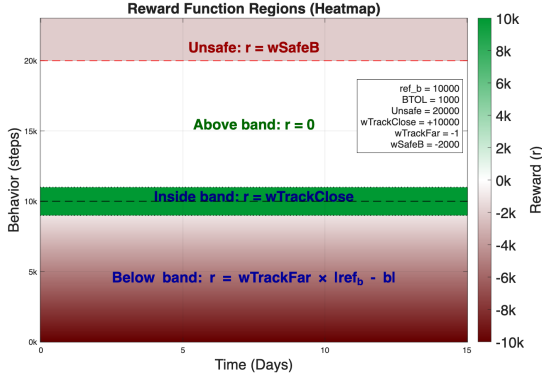


Fig. 3. Reward function structure with tracking, safety, and self-efficacy components.

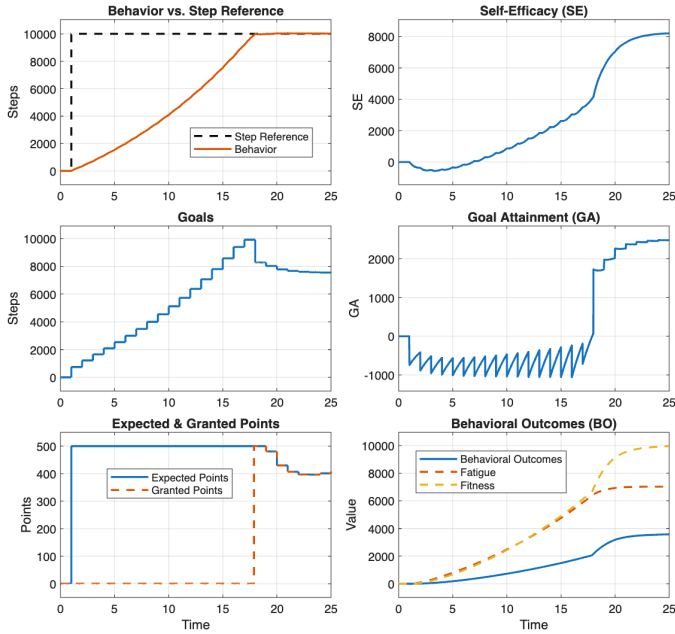


Fig. 4. Baseline MPC closed-loop simulation with fixed $\Gamma_{\Delta u_8} = 0$ and $\Gamma_{\Delta u_9} = 0$, using $u_{gr} = 500$ and the configuration in Table 1.

Fig. 4 shows a fixed-tuning MPC baseline with $\Gamma_{\Delta u_8} = 0$ and $\Gamma_{\Delta u_9} = 0$. All other MPC settings follow Table 1, with $u_{gr} = 500$. Under this configuration, behavior reaches the 10,000 steps/day setpoint only after approximately 17 days, and self-efficacy settles near 8,000 over the shown horizon. Since self-efficacy in the SCT dynamics is influenced by recent behavior and goal attainment, the achievable self-efficacy level is constrained by how quickly behavior can improve relative to the prescribed goals. This baseline motivates tuning beyond fixed move suppression and highlights the potential value of selecting horizon-dependent $\Gamma_{\Delta u_8}(1:m)$ together with an appropriate u_{gr} .

Fig. 5 shows a representative closed-loop simulation under the learned tuning after 3000 training episodes. Behavior reaches the 10,000 steps/day setpoint within 6 days of the 15-day horizon with a gradual rise, minimal overshoot, and satisfaction of the goal-attainment constraint $y_{11} \geq -1000$ throughout the episode. The resulting goal sequence changes smoothly, yielding ambitious yet feasible daily goals that support adherence. Self-efficacy also in-

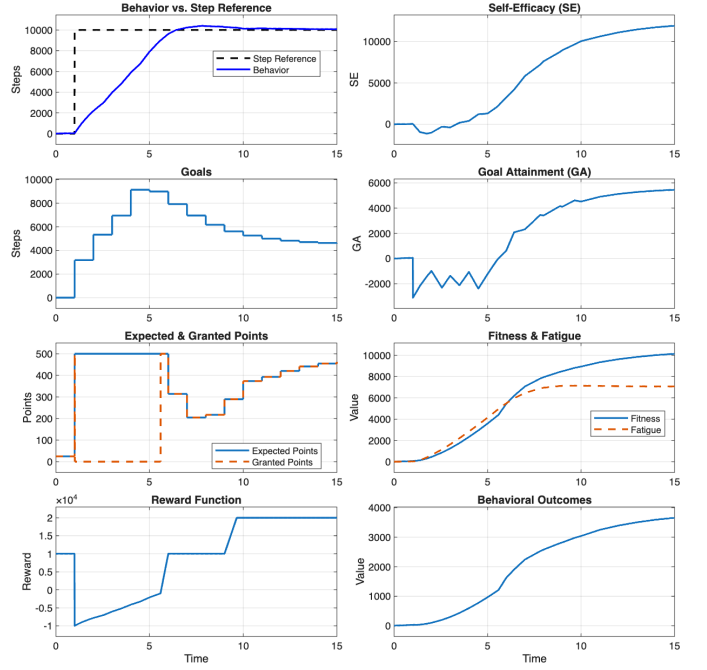


Fig. 5. Closed-loop simulation using the learned tuning parameters (Table 2) after 3000 training episodes in the Contextual bandit + MPC framework.

Table 2. Learned RL actions after 3000 training episodes for the simulation in Fig. 5.

Action	Learned values
$\Gamma_{\Delta u_8}(1:5)$	[0.4, 1000, 999.5, 999.8, 1000]
u_{gr}	500

creases substantially, reaching approximately 12,000, because the reward in Section 3.2 explicitly favors it and the learned tuning drives u_{gr} to a high value, improving goal attainment while maintaining behavior tracking. Accordingly, the reward in Fig. 5 approaches its maximum with tracking, motivation, and constraint satisfaction conditions being met, with the corresponding learned tuning parameters listed in Table 2.

Overall, the results indicate that the TD3-based tuner can efficiently search over MPC tuning parameters while MPC preserves feasibility through constraint handling. Relative to the fixed-tuning baseline, the learned horizon-dependent move-suppression profile regulates aggressiveness across the move horizon, enabling faster progress toward the step target while maintaining stable goal updates, and the learned high u_{gr} aligns with the reward emphasis on self-efficacy, suggesting that the incentive reference can support sustained engagement while maintaining tracking and constraint satisfaction. These findings support the premise that learning-based tuning can reduce reliance on manual trial-and-error selection of MPC parameters, particularly when the tuning space includes horizon-varying weights. A TD3-style actor-critic optimizer is well suited for this bandit-style tuning task because constrained QP solutions induce a nonlinear tuning-performance relationship, and its twin Q networks, deep function approximation, and off-policy replay reduce overestimation bias and improve data efficiency while yielding a parameterized tuning policy that can generalize across contexts.

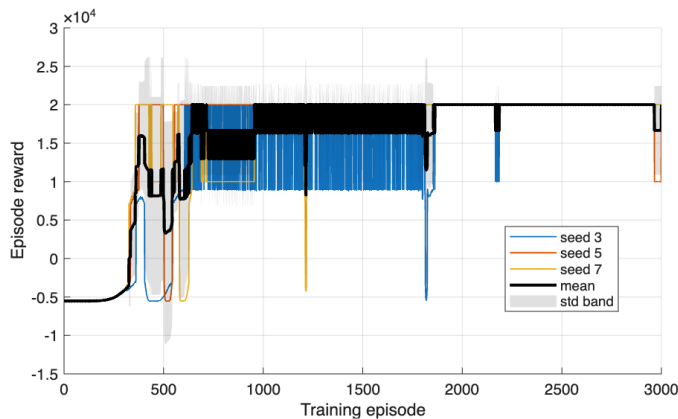


Fig. 6. Learning curves across three random seeds {3, 5, 7} (mean $\pm$ std).

To assess robustness to stochastic training effects, the TD3-style agent was trained for three random seeds, {3, 5, 7}, each for 3000 episodes. The corresponding learning curves (mean $\pm 1\sigma$ across seeds) are shown in Fig. 6. Each trained agent was then evaluated on five unseen seeds, {100, 101, 102, 103, 104}, using two evaluation episodes per seed. Across all three agents, evaluation consistently achieved the maximum terminal reward of 20000, indicating that the learned tuning policy is reproducible across training seeds and generalizes across initialization noise in the closed-loop simulation. Because all training seeds converged to the same maximum-reward solution, Table 2 reports the action values for a representative seed.

More broadly, training the contextual bandit tuner across participants with diverse trajectories can yield tuning policies that adapt MPC settings to different response patterns. The same approach can be extended to additional MPC design choices, including prediction and move horizons, input-output weights, and constraint parameters, potentially improving the tracking-motivation trade-offs while reducing computational burden by selecting efficient parameters for a given objective.

5. SUMMARY AND FUTURE WORK

This paper examines a contextual bandit with MPC framework grounded in Social Cognitive Theory in a TD3-based contextual bandit tuner selects MPC tuning parameters to generate daily step goals and expected points under a reward function that encourages tracking a 10,000 steps per day setpoint while promoting high self-efficacy as a proxy for participant engagement. The results provide preliminary evidence that learning horizon-dependent move-suppression weights and an incentive reference from closed-loop interaction data can yield smooth and feasible goal and incentive adjustments that satisfy constraints, reducing reliance on manual tuning while preserving MPC feasibility and safety.

Future work will extend the proposed contextual bandit tuned MPC framework toward stronger personalization and clinical relevance in adaptive digital health promotion. A primary direction is aligned reward design that encodes clinically meaningful priorities, behavioral sustainability, and user preferences, using alignment principles and incor-

porating expert input and participant feedback to better connect optimization objectives with intervention intent. Since contextual bandits are a one-step special case of reinforcement learning, future studies will also examine multi-step RL formulations when more frequent or event-triggered tuning updates are appropriate. In particular, reinforcement learning will be explored as a tuning mechanism for IMC style filter parameters in the 3DoF Kalman filter hybrid MPC architecture used in *YourMove* (El Mistiri et al., 2025a), enabling adaptive adjustment of closed loop time constants alongside move suppression penalties and output weighting to improve responsiveness to day to day behavioral variation and robustness to unmeasured disturbances. Generalization across participants will be pursued through transfer learning and meta reinforcement learning so that new participants can be given a warm start from shared representations and then personalized quickly with limited data.

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